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**Key Points:**
- New Viking PDS data product combines seismology and meteorology readings
- Facilitates selection of seismic data when wind is (probably) calm
- Correlation of wind and seismic data is noted, and probable dust devil is identified

**Supporting Information:**
- Supporting Information S1

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# Viking-2 Seismometer Measurements on Mars: PDS Data Archive and Meteorological Applications

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**Abstract** A data product has been generated and archived on the NASA Planetary Data System (Geosciences Node), which presents the seismometer readings of Viking Lander 2 in an easy-to-access form, for both the raw ("high rate") waveform records and the compressed ("event mode") amplitude and frequency records. In addition to the records themselves, a separate summary file for each instrument mode lists key statistics of each record together with the meteorological measurements made closest in time to the seismic record. This juxtaposition facilitates correlation of the seismometer instrument response to different meteorological conditions, or the selection of seismic data during which wind disturbances can be expected to be small. We summarize data quality issues and also discuss lander-generated seismic signals, due to operation of the sampling arm or other systems, which may be of interest for prospective missions to other bodies. We review wind-seismic correlation, the "Martian solar day (sol) 80" candidate seismic event, and identify the seismic signature of a probable dust devil vortex on sol 482 : the seismometer data allow an estimate of the peak wind, occurring between coarsely spaced meteorology measurements. We present code to generate the plots in this paper to illustrate use of the data product.

**Plain Language Summary** The only seismic data from Mars, from the Viking-2 lander, has been made available in an easy-to-access Web archive, integrated with weather data, which helps to correlate the two and to identify the signature of wind in the seismic record. This will be of use to understand data from future missions, and in fact provides new insights into Martian weather in its own right. We even use the better sampled seismic data to determine the brief peak wind in a dust devil, which occurred between Viking's meteorological measurements on the 482nd day of its mission. We also provide computer code to allow new investigators to easily explore the data.

## 1. Introduction

The Viking landers each carried a sophisticated seismometer and that of Viking 2 operated for over 500 Martian solar days (sols). Although the data are usually dismissed as of limited value due to the susceptibility of the lander-mounted instrument to wind noise, they were able to exclude high global seismicity on Mars, with only one candidate local seismic event being identified in the initial analysis (Anderson et al., 1977). The data may merit closer attention from modern geophysical data analysis methods and are in fact of interest in atmospheric science since the instrument was sampled better than the meteorology package. However, the Viking mission predates the emergence of standard formats for digital seismic data and took place before the origin of the NASA Planetary Data System (PDS) in the 1980s. Here we report an effort to make the Viking data more readily available to the community. Such a re-examination is pertinent given the forthcoming InSight mission: even a ground-emplaced and shielded seismometer (e.g., Murdoch et al., 2017; Teanby et al., 2017) will sense wind energy coupled into the ground, and possible surface deformation in response to pressure fields (e.g., Lorenz et al., 2015).

In this paper we review a data product recently delivered to the PDS. We survey lander-generated signals, which may be of interest for other planetary missions: for example, seismometers have been considered as potential payload elements for landers to Venus (Lorenz, 2012), Titan, and Europa as well as Mars.

## 2. Viking Seismometer

The Viking landers had the principal goal of detecting life on the Martian surface, and geophysical and meteorological instrumentation suffered a number of compromises in accommodation (Anderson et al., 1972, 1976, 1977). Most notably, the seismometer was mounted on the lander deck. Although for seismic

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frequencies of interest the coupling of the instrument with the ground was adequate (contrary to what is often asserted), this arrangement did cause the noise floor of the instrument to be often rather high due to wind-induced motion of the lander body on its compliant legs. For extended periods (especially at night) when wind was low, the instrument was nonetheless useful.

The instrument was a three-axis velocity sensing instrument (essentially operating as a sensitive magnet-coil geophone with an appropriate spring arrangement to give a resonant frequency of about 3 Hz). The 2.2 kg, 3.5 W instrument included variable gains and filters and (for the time) elaborate data handling.

To protect from launch, separation shocks, and landing loads, the armature of the seismometer was secured by a caging mechanism during flight. The instrument was uncaged after landing by applying a current to a release wire, which was weakened by ohmic heating. While this worked first time on Viking 2, repeated attempts on Viking 1 failed to uncage the instrument and no data were returned (Anderson et al., 1977). Some useful retrospective remarks on the seismometer experiment and its data handling are given in Lazarewicz et al. (1981).

## 3. Data Archival

Although the Viking seismometer data were in principle widely available in deep public archive at National Space Science Data Center (NSSDC), the format of that archive is not user-friendly. However, several of the original Viking investigators retained somewhat processed records. Here we use the VUSA files ("Viking Unscrambled Seismic ASCII," see Appendix A for details). These are structured as long concatenations of relatively short records, in turn forced by the limited instrument memory of the 1970s. The files include records of variable length, because different data types were interleaved as they were received, they require careful parsing and error-checking, making them cumbersome to use.

The subject of the present paper is a data set (VL2-M-SEIS-5-RDR-V1.0) now archived on the NASA Planetary Data System (PDS) Geosciences node: http://pds-geosciences.wustl.edu/missions/vlander/seismic.html. This data set distils the principal data (event and high-mode records—see later) into single ASCII files (one for each axis and type) that are readily processed with modern tools. Furthermore, since the principal obstacle to the interpretation of seismic data was the influence of wind noise, a summary product that adds the meteorological measurements made nearest in time to the seismic measurement, and notes the interval between the two, has been added. This juxtaposition first facilitates the evaluation of the seismic signature of wind (e.g., Nakamura & Anderson, 1979) or pressure or temperature changes. Second, it allows the potential influence of wind noise on candidate seismic events of interest to be assessed or permits the easy selection of records where wind was known to be weak.

The data products are described in the standard PDS documentation files ("seismic_dataset.cat," and "seismic_instr.cat"), and the file format is defined in the PDS labels associated with each file. To facilitate use of the products, we provide in the supporting information to this paper code used to generate the figures of this paper from the PDS data set above. The code is implemented in "R," a free open-source programming language and statistical analysis/graphics environment (https://www.r-project.org/).

## 4. Seismometer and Meteorological Data

The instrument (Anderson et al., 1977) had a sensitivity corresponding to about 2 nm of displacement at 3 Hz and 10 nm at 1 Hz. The instrument recorded in three modes, normal, event, and high-rate, with rates of ~6, 145, and 1,800 kbits/h, respectively. The normal mode simply records the background amplitude 4.04 times per minute and is not discussed further here. The high-rate mode samples each axis directly at 20.2 Hz, but its high data generation rate precluded extensive operation. Much of the data are in the event mode, which records the amplitude envelope in each axis at 1.01 Hz, as well as the number of positive-going zero crossings (thus a measure of frequency) in the same interval.

The archive discussed here comprises about 300,000 records, totaling about 90 million individual measurement numbers. These are not uniformly distributed with time (Figure 1): event mode data were acquired frequently during the prime mission through sol 61 (when solar conjunction limited communications with Earth: regular operations resumed on sol 89). Subsequently, seismic data were acquired in varying amounts. Data were acquired relatively consistently, however, from late at night through early morning (~23.00 h local time to about 03.00 h) when wind noise was at a minimum. High rate data were acquired much less often.

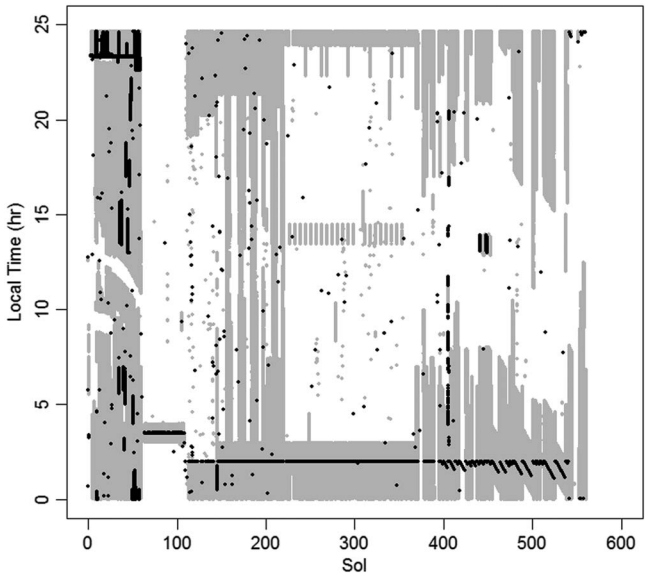



**Figure 1.** Extent (sol number versus time of day) of Viking seismometer data acquisition in high-rate (black) and event (grey) modes. Because of bandwidth limitations, relatively few high-rate measurement sessions were performed. Event mode data have a rather better coverage but are concentrated in the late night and early morning when the wind noise was minimized. The R code to generate this figure is given in the supporting information to this paper.

Measurement packets are 51 samples long in event mode, and 83 samples long in high-rate mode (to put these in a familiar modern context, it may be remarked that these record lengths are comparable with text messages or tweets). The data comprise about 49,361 high-rate records with the majority of the data set (256,087 records) in event mode. Each measurement was recorded as an integer value (for the amplitude, 7 bits) so the dynamic range is somewhat limited. Generally, the instrument was operated in its highest gain state. In total the Viking 2 seismic record corresponded to about 2100 h of measurement (89 days) up until sol ~560, when a lander mass memory unit required for seismometer data handling failed. The lander itself continued to function for much longer (~1,050 sols).

The frequency response of the seismometer was given in Figure 6 of Anderson et al. (1977). Although this diagram was given in terms of magnification based on the assumption that the digitized data were plotted at certain scales, they can be converted to a more conventional unit of digital units per ground displacement in Data Units (DU)/m by multiplying the value of magnification by 2.27, 1.96, and $1.32 \times 10^3$, for high-data-rate, event, and normal modes, respectively. For example, magnification of $10^5$ (indicated for high-rate mode for signals of 2 Hz frequency: the conversion is within a factor of 2 of this value for frequencies of 1 to ~5 Hz for all modes) translates to $2.27 \times 10^8$ DU/m or ~4.4 nm/DU. The $x$ axis is positive near-vertically upward, $+y$ is to the southwest (see Anderson et al., 1977 for full details).

The meteorology instrument on Viking included air and ground temperature sensors, a pressure sensor, and a wind speed and direction sensor. The instrumentation and results are described in Hess et al. (1977), Sutton et al. (1978), Tillman et al. (1979, 1994), and Murphy et al. (1990): see also Martínez et al. (2017). Meteorology data were not always acquired at a high rate and not always when the seismometer was operating (a total of about 1.35 million meteorology readings were taken during the first 560 sols, at intervals varying from hours down to 4 s). The summary data product archived at the PDS takes the nearest-neighbor meteorological readings to each seismic record and notes the time offset if any between them (NB this offset may be incorrectly estimated by as much as ~100 s due to different approaches to estimating local lander time from the spacecraft clock, which was found to have a small-frequency drift over the course of the mission).

## 5. Seismometer-Meteorology Comparison

The strong contribution of wind noise to the recorded data was anticipated given the nonideal accommodation of the seismometer on the lander deck (installation on a footpad would have been better) and indeed was what was observed (Anderson et al., 1976, 1977). The seismometer detects wind-generated lander motions above a wind speed of about 3 m/s, with an amplitude proportional to the square of wind speed (thus, proportional to the dynamic pressure of the wind, as would be expected; Anderson et al., 1977), and strong coherence between the wind record and the seismic noise level at a range of timescales was noted (Nakamura & Anderson, 1979) (see Figure 2).

There is a clear meteorological interpretation of the seasonal variability (see Nakamura & Anderson, 1979). In fact, the seismic variations can have meteorological interpretation on shorter timescales too—for example, the midafternoon often has very sharp pseudoperiodic variations—see Figure 3. This is the warmest part of the day when convective turbulence may be strongest, and the 3–30 min periodicity may be associated with the typical spacing of upwelling sheets in the convecting planetary boundary layer (PBL). These sheets have a spacing comparable with the PBL thickness (thus typically a few kilometers) and at wind speeds of a few meters per second are advected on ~1,000 s timescales. This ordered structure can give rise to pseudoperiodic meteorological signatures associated with dust devils (dusty whirlwinds or vertical convective vortices, which tend to form in the upwelling sheets and their corners), and indeed, regular-spaced Martian dust devils are sometimes observed (Fenton & Lorenz, 2015).



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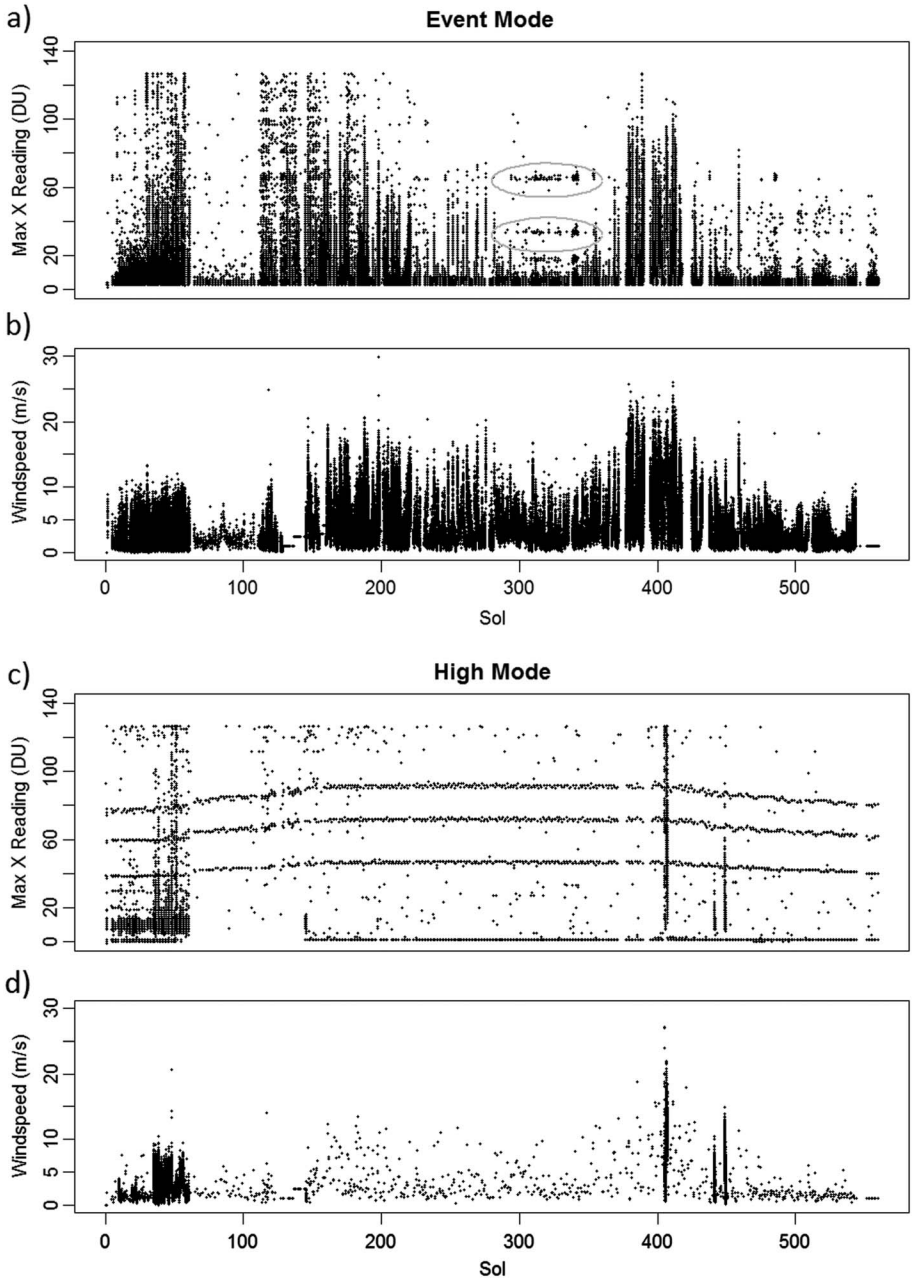



**Figure 2.** (a) The event mode maximum vertical (*x*) reading in each record throughout the mission and (b) the corresponding wind speed measurement. The correlation of the two plots is apparent—although note that here the wind speed (rather than the square of the wind, which would correlate better) is shown here. Persistent sets of data values (highlighted with grey ellipses) are calibration pulses (see later) applied to monitor instrument function. (c and d) The same but for the (much more sparse) high-rate records. The near-horizontal signature of calibration pulses in the high-rate seismic record is especially apparent—their values change slightly over the course of the mission due to changing temperatures. The vertical lines indicate that when high-rate data were abundant just after sol 400, many values were recorded.

Although the seismometer signal shows evidence of the atmospheric conditions often associated with dust devil activity, we have not noticed a distinctive signature of the individual dust devil encounters in the first 60 sols (documented by Ringrose et al., 2007). However, we present in section 6.2 a probable dust devil encounter associated with prominent seismometer signals. Note that the short-period Viking seismometer is not sensitive to the quasi-static ground tilt signatures of dust devils observed on Earth by Lorenz et al. (2015).



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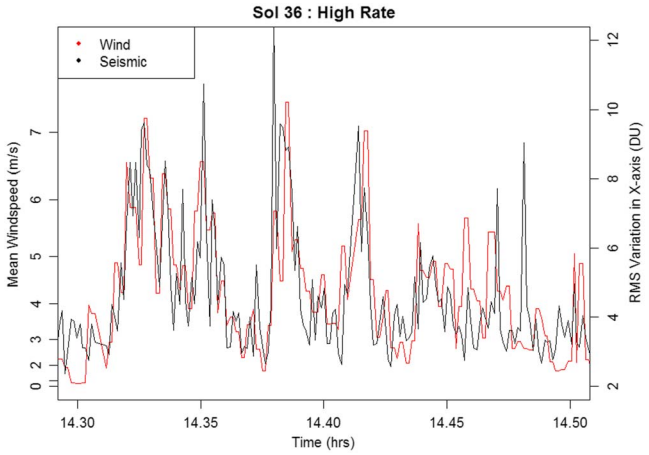



**Figure 3.** View of a high-rate measurement period on sol 36. The root-mean square variation in each *x* axis record is plotted versus time, together with the wind speed (note that the wind speed axis is nonlinear, with the *y* coordinate proportional to the square of wind speed). It is seen that the correlation is very strong (incidentally confirming the correct synchronization of the two records). It is seen that there is a pseudoperiodic character to the signal, in this instance with a dominant period of ~4 min. Inspection of the raw high-rate record (#14398 in the archive) shows that the spike at 14.48 h is due to a single outlying data point, presumably a data handling error.

We show in Figure 4 that on diurnal timescales the wind and seismic noise level are somewhat coherent, but we can detect the enhanced stability of the atmosphere at night. Despite a mean wind speed similar to earlier in the day, the seismic noise falls significantly after about 5 p.m. local time. This is presumably due to a stable layer of cold air forming by radiative cooling such that the flow past the lander is more laminar in the early morning than in more turbulent daytime conditions. Thus, it seems that the seismic noise level is diagnostic of turbulent characteristics of the wind and not purely the wind speed. Wind direction is also a factor (see Anderson et al., 1977).

## 6. Candidate Seismic Events

The general perception of the Viking seismometer investigation is that the record was contaminated by wind. However, useful constraints (i.e., upper limits) on Martian seismicity were developed from the data (e.g., Goins & Lazarewicz, 1979). Furthermore, while no large marsquake was confirmed, at least one "suspicious" event—that on sol 80—was considered to be a possible seismic event (Anderson et al., 1977). While a wind gust cannot be excluded, the assembly of meteorological data with the seismic record in our data product does support the contention that a gust origin is unlikely, as described in the next section. We also present an event in the seismic record that, despite modest wind speeds, appears to be associated with a vortex, probably a dust devil.

### 6.1. The Sol 80 Event

Measurements were made regularly in this season during the night, when wind noise was expected to be a minimum. On sol 80, at 03:00 h, a signal was captured (Figure 5) in event mode with unusual frequency characteristics, which suggested its origin as a "true" seismic event. (It is interesting that this event occurred during a period when the seismometer observation coverage was roughly at its lowest level of the mission—see Figure 1.) The signal has been compared (Anderson et al., 1977) with slightly stronger events captured at distances of 65–69 km from an identical instrument at Caltech, namely, a magnitude of 3.0 aftershock and a magnitude of 2.5 mining explosion (102 t). Assuming magnitude-duration relationships, i.e., scattering and attenuation characteristics for the Martian crust similar to Southern California, a distance of 110 km was estimated for the event and a magnitude of 2.7.

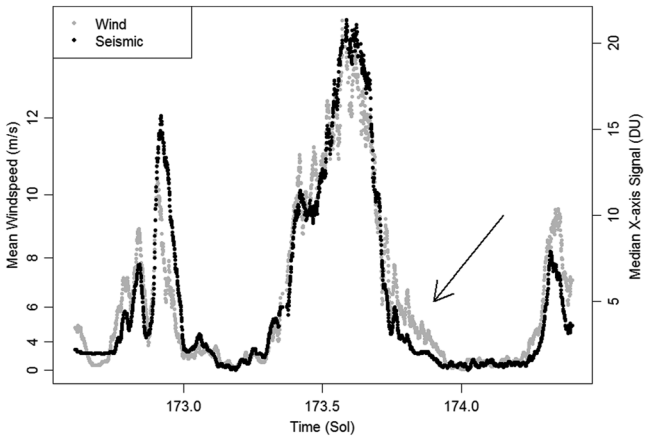



**Figure 4.** Median *x* axis seismic signal is plotted for sols 172.5–174.5, a particularly well-populated sequence of event mode records. In general, the signal tracks the wind (plotted proportional to the dynamic pressure) very well, showing a clear diurnal variation. The seismic noise in the early morning (e.g., around 173.9, arrowed) is lower compared with that seen at the same wind speed earlier in the day, at 173.4 or at 172.9.

Although the event's occurrence during the normally quiet night (when wind gusts were unlikely) was noted, gust likelihood was not quantified until the development of the present data set (Lorenz et al., 2016). Correlations of wind speed with seismometer output (Anderson et al., 1977) suggest a mean amplitude $A = 0.15\ V^2$ so the peak of 40 DU implies that a wind speed V of ~16 m/s was required to generate the lander motion shown in Figure 5. The measurements adjacent in time to the 0300 event were at 02:40 and 03:45 h, of 2.6 and 3.6 m/s (see Figure 5b). The surface winds at the Viking sites generally had a consistent diurnal cycle (see Figure 3 of Nakamura & Anderson, 1979), with wind speeds being lowest in the early hours—which is indeed why the seismometer was operating at that time. Although nighttime gusts cannot be excluded (vortex winds such as dust devils can occur at night but are much less common than in the early afternoon—e.g., Ordonez-Etxeberri et al., 2018), statistically, it seems probable that this was a true seismic event.



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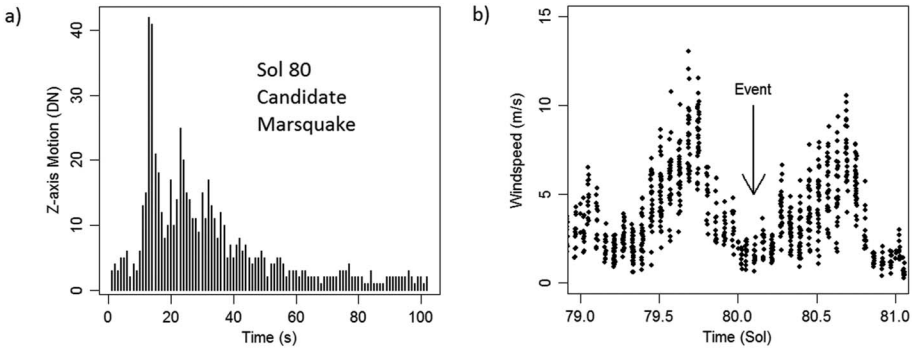



**Figure 5.** (a) *Z* axis (corresponding to a WSW-ENE direction) seismic record of the sol 80 event. Amplitudes and frequencies in all axes are presented in Anderson et al. (1977). (b) Wind data for the two days around the sol 80 event. Data were acquired in bursts at roughly hourly intervals. It is seen that an ~16 m/s wind strong enough to generate the seismic signal is somewhat improbable even during the day but is dramatically (a factor of ~4) stronger than typical for the early morning hours.

### 6.2. Sol 482 Event-Possible Dust Devil

A candidate seismic event had been identified by one of us (Y. N.) in manual inspection of the data, in the afternoon of sol 482. The new data product allows us to examine this event more closely (see Figure 6).

There is a gap in the seismic data shortly after the event, but the ~15 s event is still captured in full. Data resume ~300 s afterward, and amplitudes thereafter remain at the same background level as before the event (<5 DU). Unlike the sol 80 event, meteorological data were obtained during this period, with data acquired 30 s before and after the event. These winds are not high enough (~4.6 and 5.7 m/s, respectively) to account for the peak seismic amplitude (~26 DU). However, inspection of the wind direction in the data product shows that there is a nearly 160° change in azimuth. This abrupt veering, especially during the convectively active afternoon, is characteristic of whirlwinds such as dust devils (Lorenz, 2016; Ringrose et al., 2003, 2007). The wind speed and direction after the event return to close to what they were before.

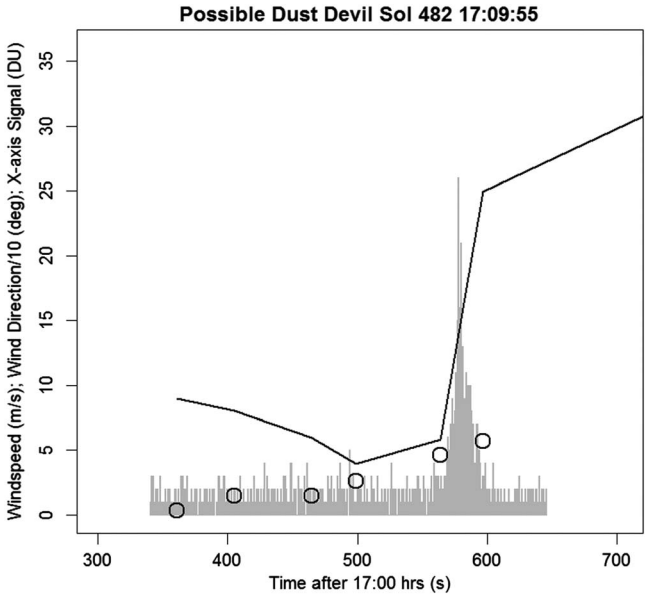



**Figure 6.** The *x* axis seismic signature (grey bars) of an event in the late afternoon of sol 482. Wind speed measurements are shown in circles, while the wind speeds measured before and after the event are not enough to account for the seismic amplitude, higher wind speeds between the measurements likely occurred, and the dramatic change in wind direction (solid line) suggests a meteorological disturbance, most probably a dust devil.

Thus, the meteorological measurements appear consistent with a close vortex passage (the vortex may or may not have been dust-laden), but they may have been too coarsely sampled to capture the actual peak turbulent wind speed encountered by the lander. If we interpret the peak seismic amplitude as indicating the maximum wind in the interval between the direct meteorological measurements, we obtain ~13 m/s. This example suggests that closer examination of the seismic record, sampled at high time resolution (compared with the meteorology package), may be fruitful in studies of boundary-later convection on Mars.

## 7. Lander-Generated Signals

Various lander mechanical activities are detectable in the seismometer data (Anderson et al., 1976, 1977), including sampling arm operations, X-ray sample dumps, tape recorder and camera operations, and S-band antenna movements. It may be noted that some of these signatures were discussed in some detail in an Appendix to Anderson et al. (1977), archived as microfiche with the hard-copy journal. That medium means the appendix is not always trivial to locate today (although at the time of writing is available online at http://authors.library.caltech.edu/54355/), and therefore, some of that discussion is repeated here. Seismic signatures of lander operations are of potential interest as a calibration source, as a crude diagnostic of activities (i.e., spacecraft health monitoring) and for outreach.



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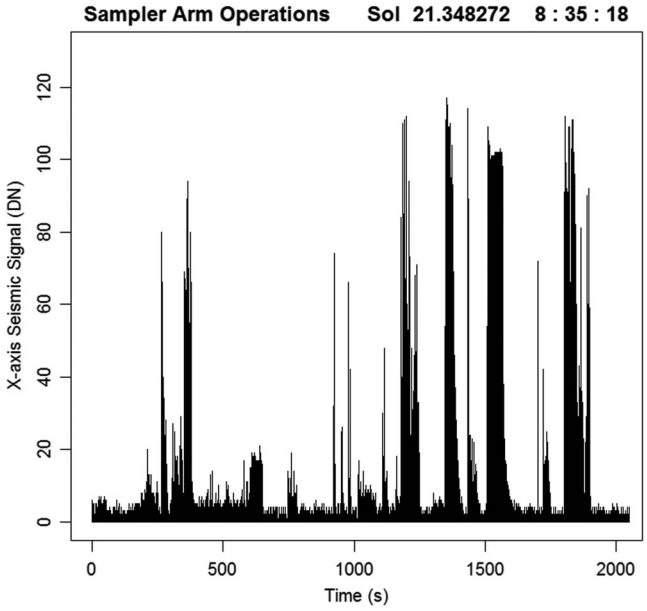



**Figure 7.** Example seismic artifact during sampling arm operations. Repeated high-amplitude signals with sudden onset (and often sudden decay) are seen, with rather low background in between.

Although complete lander operation timeline information is not available, some of the more prominent lander disturbance activities (e.g., Figure 7) are documented, e.g., some sampling arm operations (Moore, 1987), and X-ray fluorescence dumps (Baird et al., 1977). The principal operations of these systems have been noted in a summary file in the PDS archive: it may be noted that these activities were quite frequent during the first 60 sols of operations, accounting for the noisy appearance in Figure 2 for this period, despite the relatively modest wind speeds.

The (relatively infrequent) operations of the S-Band Direct-to-Earth High-Gain Antenna articulation had a distinctive signature—periodic steps about 30 s apart move the antenna to follow the Earth in the sky. An oscillation of 7.6 Hz with a $Q$ of 25 was excited by the antenna movements. The antenna mast was only 10 cm from the seismometer. Tape recorder operations produce a brief (~15 s) burst of noise.

In addition to lander mechanical noise, another nonenvironmental signal in the data is regular calibration pulses, when electromagnetic coils were used to displace the seismic mass by approximately 4 μm. The response (shown in Anderson et al., 1977, Figure 9) is a damped sinusoid, with a slightly different value for plus-minus displacements and releases: depending on which local extremum was captured in a given record, these plot as the different lines in Figure 2c. As can be seen there, and was noted in Anderson et al. (1977, p. 4532), the calibration pulse values change by about 18% over the first 140 sols (see also their Appendix A2)—the evolution is due to declining temperatures affecting the coil resistance. By sol 400, air (see, e.g., Martínez et al., 2017, Figure 5) and lander temperatures were rising again and the calibration pulse amplitude begins to decline.

## 8. Conclusion

As prospects for new seismic observations at Mars are at hand with the InSight mission, it is important that the hard-won first seismometer measurements of Viking 41 years ago be available for renewed investigation. The Viking experience shows that simultaneous high-sample rate meteorological measurements are essential to properly interpret seismometer data, since wind can vary on very short timescales, and we have presented a seismic signature associated with a probable dust devil encounter that is not prominently indicated by coarse-sampled wind speeds alone.

## Appendix A

It is useful to recall how the data set archived at the PDS was originated. The basic Viking seismic dataflow can be considered in three stages: (1) steps taken at the Jet Propulsion Laboratory (JPL) during the Viking mission, (2) steps taken at the University of Texas Institute for Geophysics (UTIG) during and shortly following the Viking mission, and (3) steps taken at UTIG recently.

In stage (1), the bit streams of seismic data as received from Mars were recorded on seven-track reel-to-reel tapes called SEISF tapes. Copies of SEISF tapes were sent to UTIG for their use. At JPL, SEISF tapes were reformatted further to generate SEISS and SEISW tapes, which were supposed to have been archived, but their whereabout is unknown at this time. SEISW tapes were further reformatted to generate Experiment Data Record (EDR) tapes, which were sent to Massachusetts Institute for Technology for their use, and their copies were sent to NSSDC for archiving. (Files read from EDR tapes were reformatted to create EDR-2 files, which are now available from NSSDC.)

In stage (2), the data on SEISF tapes were reformatted (UnScrambled) to create Viking Unscrambled Seismic (VUS) tapes, and these tapes were used for their analyses during the mission. Later on, in the early 1990s, the data on VUS tapes as well as those on SEISF tapes were copied to an 8 mm cassette



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tape, a copy of which was sent to Incorporated Research Institutions for Seismology (IRIS) for archiving and another copy was sent to Institute of Space and Aeronautical Science, the predecessor of a division of the Japanese Aerospace Agency (JAXA), for their use. NSSDC recently obtained a copy of files on the cassette tape from IRIS to add to their archives. In stage (3), the data on VUS cassette tapes were further decoded and converted to ASCII format to create VUSA files. These Viking Unwrapped (Unscrambled) Seismic files in ASCII have been maintained by one of us (Y. N.) and are presently available at ftp://ftp.ig.utexas.edu/pub/Viking/

An archive of both the EDR-2 files from NSSDC, and the VUS tapes from UTIG, is presently available at the Japanese Aerospace Agency (JAXA) website

https://darts.jaxa.jp/planet/seismology/viking/index.html

It should be noted that every reformatting step had a chance to introduce errors, and the Viking seismic data sets are no exception.

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